

Math Camp

Lecture 10

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Second-order conditions

- As with the unconstrained problem, finding a critical point does not imply the point is a local maximum
 - It could be a local minimum or a saddle point
- Second-order conditions sufficient to guarantee that a critical point is a local maximum are available
 - But they require a modified version of the Hessian to account for the presence of the constraints

The bordered Hessian

- Let $\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda'(\mathbf{c} - g(\mathbf{x}))$. Then

$$D^2 \mathcal{L}(\mathbf{x}, \lambda) = \begin{pmatrix} D_{\mathbf{x}}^2 \mathcal{L} & -Dg' \\ -Dg & 0 \end{pmatrix}$$

- ▶ $D_{\mathbf{x}}^2 \mathcal{L}$ is the matrix of cross partials of $\mathcal{L}$ with respect to $\mathbf{x}$
- ▶ Dg is the derivative of g with respect to $\mathbf{x}$
- ▶ Because λ enters the Lagrangian linearly, $\mathcal{L}_{\lambda\lambda} = 0$
- $D^2 \mathcal{L}(\mathbf{x}, \lambda)$ is called the *bordered Hessian*
 - ▶ The Hessian of $\mathcal{L}$ with respect to $\mathbf{x}$, bordered by the derivatives of the constraint

Second-order conditions

Theorem

If $(\mathbf{x}^*, \lambda^*)$ satisfies Lagrange's FOC and if $\mathbf{v}' D_{\mathbf{x}}^2 \mathcal{L}(\mathbf{x}^*, \lambda^*) \mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\ker Dg(\mathbf{x}^*)$, then $\mathbf{x}^*$ is a local maximum.

- $D_{\mathbf{x}}^2 \mathcal{L}$ need not be negative definite everywhere, only on the kernel of Dg
 - $\ker Dg(\mathbf{x}^*)$ is exactly the set of variations that keep the constraint satisfied
 - Those are the only variations available to us, so they are the only ones the second-order condition has to rule on

The envelope theorem

- The envelope theorem allows for a simple assessment of how the *value* of the maximization problem changes when a parameter changes
- Begin with $n = 2$ and $m = 1$. Write

$$V(c) = \max_{\mathbf{x} \in K} f(\mathbf{x})$$

where K is defined by $g(\mathbf{x}) = c$

- The value function V depends on the parameter controlling the constraint c
- We want to compute $V'(c)$

The envelope theorem

- Note that

$$g_1 \, dx_1 + g_2 \, dx_2 = dc \quad \text{or} \quad dx_2 = \frac{1}{g_2}(dc - g_1 \, dx_1)$$

- Also $dV = f_1 \, dx_1 + f_2 \, dx_2$. Combining,

$$dV = f_1 \, dx_1 + \frac{f_2}{g_2}(dc - g_1 \, dx_1) = \left(f_1 - \frac{g_1}{g_2} f_2\right) dx_1 + \frac{f_2}{g_2} dc = \lambda \, dc$$

- It follows that $V'(c) = \lambda = \mathcal{L}_c$
 - To compute $V'(c)$, just differentiate the Lagrangian

The envelope theorem

- It also provides an intuition for the multiplier
 - λ measures the marginal value obtained by relaxing the constraint
 - It's the price of the constraint
- In economics it is usually called the *shadow price* of the constraint
 - The marginal benefit of loosening the constraint by one unit
 - It is a price even though nobody posts it

The envelope theorem

Envelope Theorem

Consider the maximization problem

$$V(\eta) = \max_{\mathbf{x} \in K(\eta)} f(\mathbf{x}, \eta)$$

where, $\eta \in \mathbb{R}^k$ and $K(\eta) = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}, \eta) = 0\}$

If $(\mathbf{x}^*, \lambda^*)$ satisfy the Lagrange conditions and constitute a local maximum then

$$V_{\eta_i}(\eta) = \mathcal{L}_{\eta_i}(\mathbf{x}^*, \lambda^*)$$

Complementary slackness

- To handle inequality constraints we need *complementary slackness*, which pertains to pairs of weak inequalities
- We say that $x \geq 0$ and $y \geq 0$ hold with complementary slackness provided that $xy = 0$
 - Exactly the same terminology is used for the other three possible permutations of inequality pairs
 - $x \geq 0$ and $y \leq 0$, $x \leq 0$ and $y \geq 0$ or, $x \leq 0$ and $y \leq 0$.
- Intuitively it says that if one inequality is strict then the other isn't

Inequality constraints

- Inequality constraints present similarly to equality constraints
- Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{c} \in \mathbb{R}^m$. Then

$$K = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) \leq \mathbf{c}\}$$

where the suggestive notation means $g^i(\mathbf{x}) \leq c_i$ for $i = 1, \dots, m$

- Again we could always rewrite g to set $\mathbf{c} = \mathbf{0}$, but it is often convenient in applications to have non-zero $\mathbf{c}$
- By far the most common inequality constraints in economics are the *non-negativity constraints* $x_1 \geq 0, \dots, x_n \geq 0$

Inequality constraints

- For intuition consider again $n = 2$ and $m = 1$, and assume $\mathbf{x}^*$ is a local maximum
- If the constraint is *not binding*, i.e. $g(\mathbf{x}^*) < c$, then it is as if the constraint doesn't even exist
 - The FOC are the same as in the unconstrained case, $f_i(\mathbf{x}^*) = 0$
- If the constraint is *binding* then $g(\mathbf{x}^*) = c$ and we are back in the equality-constraint case
 - Under the language of Lagrange, $f_i(\mathbf{x}^*) - \lambda^* g_i(\mathbf{x}^*) = 0$

Inequality constraints

- Another point in the binding case: $\lambda^* \geq 0$
 - If it were negative then tightening the constraint would increase the value of the objective
 - But this means we shouldn't have been on the constraint in the first place

Inequality constraints

- Notice that the unconstrained FOC is reproduced by the constrained FOC when $\lambda^* = 0$
- This observation, together with the language of complementary slackness and the non-negativity of λ^* , allows us to succinctly combine the two cases

$$f_i(\mathbf{x}^*) - \lambda^* g_i(\mathbf{x}^*) = 0, \quad \text{and} \quad (\lambda^* \geq 0 \text{ and } g(\mathbf{x}^*) \leq c \text{ hold w/cs})$$

- Even more succinctly, in terms of the Lagrangian, the FOC are

$$\mathcal{L}_i = 0 \quad \text{and} \quad (\lambda^* \geq 0, \mathcal{L}_\lambda \geq 0 \text{ w/cs})$$

Kuhn–Tucker

- Now we are ready to put this all together
- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be C^2
- The general finite-dimensional constrained problem may be written

$$\max_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

$$g^i(\mathbf{x}) = c_i, \quad i \in \{1, \dots, m\}$$

$$h^j(\mathbf{x}) \leq 0, \quad j \in \{1, \dots, k\}$$

- Let $\mathcal{L} : \mathbb{R}^n \oplus \mathbb{R}^m \oplus \mathbb{R}^k \rightarrow \mathbb{R}$ be given by

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = f(\mathbf{x}) + \lambda'(\mathbf{c} - g(\mathbf{x})) - \mu' h(\mathbf{x})$$

Kuhn–Tucker

- A point $\mathbf{x} \in \mathbb{R}^n$ satisfies the *constraint qualification* (CQ) if the derivatives of the equality constraints g^i , and of the inequality constraints h^j for which the constraint is binding, i.e. $h^j(\mathbf{x}) = 0$, are linearly independent

Kuhn–Tucker

Kuhn–Tucker Theorem

If $\mathbf{x}^*$ solves the problem and satisfies (CQ) then there exist $\lambda^* \in \mathbb{R}^m$ and $\mu^* \in \mathbb{R}^k$ so that

- $\mathcal{L}_{x_p}(\mathbf{x}^*, \lambda^*, \mu^*) = 0, p \in \{1, \dots, n\}$
- $\mathcal{L}_{\lambda_i}(\mathbf{x}^*, \lambda^*, \mu^*) = 0, i \in \{1, \dots, m\}$
- $\mathcal{L}_{\mu_j}(\mathbf{x}^*, \lambda^*, \mu^*) \geq 0$ and $\mu_j^* \geq 0$, w/cs $j \in \{1, \dots, k\}$

Kuhn–Tucker

$$\max_{x_1 \geq 0, x_2 \geq 0} -\left((x_1 - 4)^2 + (x_2 - 4)^2\right)$$

subject to,

$$x_1 + x_2 = 4$$

$$x_1 + 3x_2 \leq 9$$

Corner solutions

- Interior solutions are found with the conditions we already have; corner solutions sit on the boundary of K
 - Both can occur in the same problem, and both must be checked
- In economics we usually optimize over the non-negative orthant, so the corners are the points at which some good is not consumed at all

Corner solutions

- Consider a problem with a “good” and a “bad”
 - Consuming less of the bad is always better
 - But we cannot consume a negative amount, so the solution is to consume none of it
- This is the case in which the non-negativity constraint binds and its multiplier is strictly positive
 - The unconstrained FOC would have called for $x^* < 0$
 - The constraint stops us at $x^* = 0$, and the multiplier measures how much we would gain if it did not